MIL-E-22200/10B(SH)
24 March 1989
SUPERSEDING
MIL-E-0022200/10A(SH)
8 July 1983
(See 6.7)

MILITARY SPECIFICATION

ELECTRODES, WELDING, MINERAL COVERED, IRON-POWDER, LOW-HYDROGEN MEDIUM, HIGH TENSILE AND HIGHER-STRENGTH LOW ALLOY STEELS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers iron-powder, low-hydrogen types of covered electrodes for the fabrication and repair welding of low and medium carbon, high tensile, and higher-strength low-alloy steels for as-welded and stress-relieved application.

1.2 Classification. Iron-powder, low-hydrogen electrodes shall be furnished in the types, classes, and sizes shown in table I, as specified (see 6.2).

TABLE I. Types, classes, and sizes.

Types[1/ 2/]	Class	Positions of welding	Sizes[3/] (diameter) (inches)
MIL-7018-M	1	All	3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 7/32, 1/4
	3	Flat	5/16
MIL-10018-M1 MIL-12018-M2	1	All	3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 7/32, 1/4

See footnotes at top of next page.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1/ Type classifications of electrodes shall be determined according to the capability of the electrode to meet the tests specified herein, when used in the positions of welding specified in table I and in section 4. An electrode will be approved under one type classification and an electrode will not be approved if it differs in type from that stated in the manufacturer's request for test. The position description indicates the maximum size qualified under the specification. However, reference should be made to the Qualified Products List (QPL) for the selection of electrodes for all positions of welding.

2/ The suffix RC added to the MIL type pertains to restriction of the copper content (see 3.5). Requirements, tests, and acceptance criteria remain otherwise unchanged.

3/ Electrodes of other sizes or for other positions may be tested for qualification subject to NAVSEA approval.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-S-16216 - Steel Plate, Alloy, Structural, High Yield Strength (HY-80 and HY-100).
MIL-E-22200 - Electrodes, Welding, Covered; General Specification for.
MIL-S-22698 - Steel Plate and Shapes, Weldable Ordinary Strength Higher Strength: Hull Structural.
MIL-S-24645 - Steel, Plate, Sheet, or Coil, Age-Hardening Alloy, Structural, High Yield Strength (HSLA-80).

STANDARDS

MILITARY

MIL-STD-271 - Requirements for Nondestructive Testing Methods.
MIL-STD-2149 - Standard Procedures for Explosion Testing Ferrous and Non-Ferrous Metallic Materials and Weldments.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (Attn: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.1.2 Other Government publication. The following other Government publication forms a part of this document to the extent specified herein. Unless otherwise specified, the issue is that cited in the solicitation.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
0900-LP-003-9000 - Radiographic Standards for Production and Repair Welds.

(Application for copies should be addressed to the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A 36 - Standard Specification for Structural Steel.
A 131 - Standard Specification for Structural Steel for Ships.
A 285 - Standard Specification for Pressure Vessel Plates, Carbon Steel, Low- and Intermediate-Tensile Strength. (DoD adopted)
A 710 - Standard Specification for Low-Carbon Age-Hardening Nickel-Copper-Columbium Alloy Steels.
E 604 - Standard Test Method for Dynamic Tear Testing of Metallic Materials. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)
A4.3 - Standard Methods for the Determination of Diffusible Hydrogen content of Martensitic, Bainitic, and Ferritic Steel Weld Metal Produced by Arc Welding.
B4.0 - Standard Methods for Mechanical Testing of Welds. (DoD adopted)

(Application for copies should be addressed to the American Welding Society, Inc., 550 NW LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

(Non-government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Electrodes furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.3 and 6.4).

3.2 General. Electrodes furnished under this specification shall conform to the requirements of MIL-E-22200 and as specified herein.

3.3 Composition and iron content.

3.3.1 Composition. The chemical composition of the electrode covering, except as specified in 3.3.2, is optional with the contractor. The water content of the covering shall not exceed the percent-by-weight values specified in table II. In addition, the diffusible hydrogen levels in milliliters per one hundred grams (mL/100g) of deposited weld metal shall not exceed the following values for welds deposited with electrodes removed from sealed manufacturer's containers (see 4.6):

(a) Electrodes 1/8-inch size and smaller, maximum average value of 3.2 mL/100g and maximum single value of 4.0 mL/100g.
(b) Electrodes 5/32-inch size and larger, maximum average value of 3.5 mL/100g and maximum single value of 4.3 mL/100g.

TABLE II. Water content.

Types	Percent by weight maximum	
	Condition A1/	Condition B2/3/
MIL-7018-M	0.10	0.20
MIL-10018-M1	.10	.20
MIL-12018-M2	.10	.20

1/ Condition A - When removed from sealed manufacturer container.
2/ Condition B - After 9 hours exposure at 80 percent relative humidity (R.H.) and 80 degrees Fahrenheit (°F) (see 4.7).
3/ For receipt inspection by the procuring activity, the maximum value shall be 0.25.

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3.3.2 Total iron content. The total iron content of the covering, including any combined iron in addition to metallic iron powder, shall be not less than 15 percent (see table VI).

3.4 Groove welds. The mechanical properties of groove welds shall be as specified in table III. When specified (see 6.2), the applicable stress relieving treatments shall be as follows:

(a) MIL-7018M: 1,150 ± 25°F, held at this temperature for 1 hour minimum and furnace cooled at 200°F per hour or slower to below 500°F.
(b) MIL-10018-M1: 1,125 ± 25°F, held at this temperature for 1 hour minimum and furnace cooled at 200°F per hour or slower to 500°F.

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TABLE III. Mechanical properties.[1/]

Types	Yield strength[2/]		Elongation percent in 2 inches		Transverse side bend	Impact Tests						
						Charpy V-notch impact[5/]				Dynamic tear[6/] (As-welded)		Explosion Test
	As-welded[3/]	Stress-relieved	As-welded	Stress-relieved	As-welded	As-welded		Stress-relieved				
	lb/in^2/	lb/in^2/ min	min	min		Energy ft-lb min average	Temperature °F	Energy ft-lb min average	Temperature °F	Energy ft-lb min	Temperature °F	As-welded
MIL-7018-M	53,000 to 72,000	51,000	24	24	---	50	-20	35	-20	---	---	---
MIL-10018-M1	82,000 to 100,000	80,000	20	20	4/	35	-60	20	-60	300	-20	7/
						60	0	50	0	450	30	
MIL-12018-M2	102,000 to 120,000	---	18	---	4/	45	-60	---	---	400	-20	7/
						60	0			575	30	

1/ The ultimate tensile strength and percent reduction of area shall be recorded.
2/ Yield strength shall be determined by the offset method (offset equals 0.2 percent), and shall be the average of two tests with no value more than 2 ksi below the minimum specified value.
3/ The upper value of the yield strength may be 5,000 pounds square inch (lb/in^2) higher than the indicated value for the 3/32-inch size electrode.
4/ After bending, the convex surface of the specimen shall have no visual cracks exceeding 1/8 inch. The corners of the specimen shall have no visual cracks exceeding 3/16 inch. The bend radius shall be two times the plate thickness (2t), and the tests conducted in accordance with AWS B4.0.
5/ For each testing temperature, the average values of five tests shall be greater than the minimum average value specified. No two specimens shall have values below the minimum average specified. One specimen may have a value of 10 foot-pounds below the minimum average specified.
6/ Requirement for 5/8 inch dynamic tear test shall be in accordance with ASTM E 604 (see 4.5.1). Where both Charpy V-notch and dynamic tear testing are performed, the dynamic tear test shall be considered the authoritative test.
7/ The explosion test series shall be fabricated and tested (see 4.5.2). The criteria of performance for two crack-starter weldments shall be as follows:

(a) First shot - Crack-starter bead shall crack. If the crack-starter bead does not crack, the first shot shall be repeated. Percent reduction in thickness shall be recorded. No piece shall be thrown out of weldments. No through-thickness cracks shall be present. No crack shall extend into hold-down area (that area outside a 12-1/2-inch radius circle measured from the weldment center.)
(b) Second shot - Percent reduction in thickness shall be recorded. No piece shall be thrown out of weldment. Through-thickness cracks are acceptable. No cracks shall extend into hold-down area (that area outside a 12-1/2-inch radius circle measured from the weldment center.)

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3.5 Chemical composition of deposited weld metal. Chemical composition of deposited weld metal shall be as specified in table IV.

TABLE IV. Chemical composition (percent) 1/ of deposited weld metal. 2/

Element	MIL-7018-M	MIL-10018-M1	MIL-12018-M2
Carbon	0.12	0.10	0.10
Manganese	.40 to 1.60	.80 to 1.85	.80 to 1.85
Silicon	.80	.65	.65
Phosphorus	.030	.025	.025
Sulfur	.020	.017	.012
Chromium	.15	.40	.65
Nickel	.25	1.25 to 3.00	1.50 to 4.00
Molybdenum	.35	.50	.90
Vanadium	.05	.05	.05

1/ Single values are maximum percentages.
2/ When specified (see 6.2), use of the suffix RC, for example MIL-10018-M1 RC, indicates that the maximum copper content of the deposited weld metal shall be 0.15 percent. The remaining basic compositional requirements remain unchanged for the specific MIL-type.

3.6 Electrode identification. Electrodes shall be identified as specified in MIL-E-22200.

3.6.1 Heat and lot identification. When specified (see 6.2), each electrode shall be marked with heat lot or other controlled marking code identification as specified in MIL-E-22200.

3.7 Electrode lengths. The lengths of electrodes shall be as specified in table V.

TABLE V. Electrode lengths.

Diameter (inch)	Electrode lengths overall (inches)
3/32	9, 12, or 14 1/
1/8, 5/32, 3/16	14
7/32, 1/4, 5/16	18

1/ (See 6.2)

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3.8 Core wire alloy identify. The core wire alloy identity requirements shall be as specified in MIL-E-222200 (See table VI).

3.9 Soundness. Electrodes shall deposit weld metal which meets the nondestructive requirements as specified in radiographic inspection-class 1 of NAVSEA 0900-LP-003-9000 (see 4.5.1(k)).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and compliance. The responsibility for inspection and compliance shall be in accordance with MIL-E-22200 and as specified herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance (see 4.4).

4.3 Qualification tests. Qualification tests shall be in accordance with MIL-E-22200 and tables VI and VII herein. Schedule A tests shall be conducted by the contractor, and upon successful completion of schedule A tests, schedule B welding and testing will be conducted by the Government. Schedule B test weldment shall be in accordance with 4.5.2.1.

4.3.1 Schedule A testing. Schedule A testing shall be required for all electrodes, and manufacturer certified test data shall include:

(a) Data acquired from a production run sampling.
(b) Material traceable from the heat batch through coupon welding to the test specimen.
(c) Coupon welding parameters and bead placement.
(d) Test specimen data as required by table VII and supplemented by adjunct table VI testing.

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TABLE VI. Summary of other tests.

Tests	Procedures	Requirements
Visual and dimensional	MIL-E-222200	MIL-E-222200 and 3.7 herein
Concentricity		MIL-E-22200
Dielectric strength 1/		MIL-E-22200
Covering, flaking, and cracking 1/		MIL-E-22200
Total iron in covering 1/		3.3.2 herein
Alloy identity for core wire		MIL-E-22200
Covering moisture	MIL-E-22200 and 4.7 herein	3.3.1 herein
Diffusible hydrogen	4.6 herein	3.3.1 herein

1/ Required for qualification (QPL) only.

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Table VII. Summary of weld metal tests required for qualification.[1/][2/]

Types	Test schedule	Chemical	Radiograph	Tensile	Bend	Impact[3/]		Explosion test[4/]	Test procedures	Requirements
						Charpy V-notch	Dynamic tear			
MIL-7018-M	A	X							MIL-E-22200	Table IV herein
			X	X		X			4.5.1 and figure 1 herein	3.9 herein and Table III herein
MIL-10018-M1 and MIL-12018-M2	A	X							MIL-E-22200	Table IV herein
			X	X		X			4.5.1 and and figure 1 herein	3.9 herein and Table III herein
							X		4.5.1 and figure 1 herein and ASTM E 604	Table III herein
	B							X	MIL-STD-2149 and 4.5.2 herein	Table III herein
			X						4.5.1 herein	3.9 herein
				X	X	X			4.5.1 herein	Table III herein
							X		4.5.1 herein and ASTM E 604	Table III herein

1/ Tests shall be conducted for both the as-welded and stress-relieved conditions as applicable.
2/ Electrode types shall be tested using direct current (dc), reverse polarity.
3/ For the types MIL-10018-M1 and MIL-12018-M2 electrodes, both the Charpy V-notch and the dynamic tear test shall be conducted for conformance to table III.
4/ For the types MIL-10018-M1 and MIL-12018-M2 electrodes, all sizes shall be qualification tested in accordance with MIL-STD-2149 for conformance to table III herein, except a smaller size electrode of the same type and class previously qualified shall be evaluated on the basis of Charpy V-notch and dynamic tear tests conducted at table III temperatures and the explosion test assemblies shall be omitted.

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4.3.2 Schedule B testing. Schedule B testing shall be required for MIL-10018-M1 and MIL-12018-M2 only, and test data shall be presented in an Engineering Technical Report which includes:

(a) Proof test data substantiating schedule A data.
(b) Explosion test result evaluation.

4.4 Quality conformance inspection.

4.4.1 Quality conformance inspection. For type MIL-12018-M2, quality conformance tests shall be conducted on samples selected as specified in 4.4.3 from electrodes produced from each lot as specified in 4.4.2. One sample shall be subjected to all the tests specified in tables VI and VIII herein (that is, radiograph, mechanical, and complete chemical analysis). For types MIL-7018-M and MIL-10018-M1, quality conformance tests shall be conducted on samples selected as specified in 4.4.3 from electrodes produced from each wet mix of covering mixture in each lot as specified in 4.4.2. One sample shall be subjected to all the tests specified in tables VI and VIII herein (that is, radiograph, mechanical, and complete chemical analysis). The samples representing each of the remaining wet mixes of covering mixture in each lot shall be subjected to equivalency tests as specified in 4.4.1.1.

TABLE VIII. Summary of weld metal tests required for quality conformance inspection.

Types[1/2/]	Chemical	Radio-graph	Tensile	Impact[1/]		Test procedures	Requirements
				Charpy V-notch	Dynamic tear		
MIL-7018-M	X					MIL-E-22200	Table IV herein
MIL-10018-M1 and MIL-12018-M2		X	X	X		4.5.1 and figure 1 herein	3.9 Table III herein
MIL-10018-M1 and MIL-12018-M2					X	4.5.1, figure 1 herein and ASTM E 604	Table III herein

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1/ For type MIL-10018-M1 and MIL-12018-M2, either the Charpy V-notch or the dynamic tear test shall be conducted for conformance to the impact requirements as specified in table III.
2/ For type MIL-7018-M and MIL-10018-M1, stress relieved mechanical and impact properties shall be determined only when specified (see 6.2).

4.4.1.1 Wet mix equivalency tests. Weld deposits shall be chemically analyzed for manganese, nickel (except MIL-7018-M) and copper (for MIL-types with and RC suffix) using procedures specified in MIL-E-22200 to the requirements of Table IV. Wet mix equivalency tests are not needed when the lot is identified by controlled covering mixture chemical composition as defined in MIL-E-22200. The alternate chemical analysis method of MIL-E-22200 may be used for these tests.

4.4.2 Lot. A lot of electrodes shall be as specified in MIL-E-22200 for inspection level B for types MIL-7018-M and MIL-10018-M1. A lot of electrodes shall be as specified in MIL-E-22200 for inspection level A for type MIL-12018-M2.

4.4.3 Sampling for quality conformance inspection. Sampling for quality conformance inspection shall be in accordance with MIL-E-22200, except that electrodes for the covering moisture or the diffusible hydrogen test taken from the production line shall be packaged for delivery to the test site. Any packages taken from stock shall be unopened when delivered to the test site.

4.4.3.1 Unsatisfactory test results. If the results of the first tests are determined to be unsatisfactory, two retests of the failed test shall be permitted. The results of both retests shall meet the requirements for quality conformance inspection.

4.4.4 Visual and dimensional inspection. Before the quality conformance inspection has begun, a visual and dimensional inspection shall be made in accordance with MIL-E-22200.

4.5 Weld metal test procedures.

4.5.1 Flat and vertical groove-weld metal test. The dimensions of the test weldment and the location of the tensile and impact specimens are shown on figure 1. The test assembly shall be welded, machined, and tested as specified below. Tests conducted shall be in accordance with schedule A of table VII for qualification or table VIII for quality conformance inspection. The test weldment (see figure 1) may be varied in length to suit the number of specimens that are required.

(a) For test specimens of weld metal deposited by the type MIL-7018-M electrodes, steel base plate materials (see figure 1) shall be in accordance with either MIL-S-22698, ASTM A 36, ASTM A 131, or ASTM A 285, grade A or B.

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(b) For weld metal deposited by type MIL-10018-M1 electrodes, the base plate material shall be HY-80 steel in accordance with MIL-S-16216 when dynamic tear tests are being conducted. When Charpy V-notch tests are being conducted, the base plate material shall be HY-80 steel in accordance with the chemical composition requirements of MIL-S-16216. The plate material can be in the as-rolled condition without being quenched and tempered nor tested for mechanical properties. When specified (see 6.2), a second test weldment shall be made and quality conformance tested using ASTM A 710 grade A or MIL-S-24645 grade HSLA-80 steel base plate material.

(c) For weld metal deposited by type MIL-12018-M2 electrodes, the base plate material shall be HY-100 steel in accordance with MIL-S-16216 when dynamic tear tests are being conducted. When Charpy V-notch tests are being conducted, the base plate material shall be HY-100 steel in accordance with the chemical composition requirements of MIL-S-16216. The plate material can be in the as-rolled condition without being quenched and tempered nor tested for mechanical properties.

(d) For type MIL-10018-M1 and MIL-12018-M2 electrodes, the welding position, welding current, preheat and interpass temperatures, and heat input, shall be as specified in table IX. The backing bar shall be 1/4 inch thick for the flat position tests and 1/2 inch thick for the vertical-up position tests. The weld reinforcement shall be no greater than 3/16 inch. The test plates shall be insulated or separated from the work bench. Tensile and impact tests shall be performed on both flat and vertical-up position weldments (see figure 1). Welding current shall be d.c., reverse polarity.

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TABLE IX. Cooling rate parameters for MIL-10018-M1 and MIL-12018-M2 test weldments.

Welding position	Electrode diameter (inches)	Welding current +/- 5% (amperes)	Plate thickness (inch)	Preheat and interpass temperature (°F)1/	Heat input (kilojoules/inch)	
					MIL-10018	MIL-12018
Flat	3/32	95	3/4	275-300	30 to 40	30 to 40
	1/8	140			45 to 55	40 to 50
	5/32	170			50 to 60	50 to 60
	3/16	225				
	7/32	300				
	1/4	350				
	5/16	430				
Vertical-up	3/32	95	3/4	200-250	25 to 35	30 to 40
	1/8	125			40 to 50	50 to 60
	5/32	145				

1/ Minimum preheat for ASTM A 710 steel and MIL-S-24645 shall be 60°F.

(e) For type MIL-7018-M electrodes, one test weldment shall be fabricated using the flat position with 3/16 inch and larger diameter electrodes and the vertical-up position with 5/32 inch and smaller diameter electrodes. The preheat and interpass temperature shall be 225 ± 25°F except for ASTM A 710 steel and MIL-S-24645 which shall be 60°F minimum. The heat input shall be 50,000 to 60,000 joules per inch for the 1/8 inch and larger diameter electrodes and 30,000 and 40,000 joules per inch for the 3/32 inch diameter electrode. The weld reinforcement shall be not greater than 3/16 inch. The backing bar shall be 1/2 inch thick.

(f) Heat input requirements for test assemblies:

(1) Operational heat input shall be the actual heat input recorded or calculated for individual passes during welding of the test assemblies. No more than 20 percent of the individual passes shall fall outside the minimum and maximum of the range specified.

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(2) Average heat input shall be the average of the operational heat input of all passes in the test assembly and shall fall at the mid-point of the range specified plus or minus 2,000 joules per inch.
(3) The root pass may be excluded from all heat input calculations.
(4) A weld pass shall be defined as one bead extending from the beginning to the end of the test assembly. Each weld pass shall be completed prior to commencing any other weld pass.

(g) For MIL-7018-M welding amperage shall be in accordance with the manufacture's recommendations and the actual values used shall be recorded and reported. Welding current shall be d.c., reverse polarity.
(h) Peening of weld beads shall not be permitted.
(i) Welding may be continuous except for interpass cooling. No post weld heat treatment shall be employed except for stress relief, where required, of MIL-7018-M and MIL-10018-M1 weldments. For test plates of HY-80 and HY-100, all time delay per pass beyond that necessary for interpass cooling shall be reported.
(j) Weld bead placement shall be as shown on figure 2.
(k) For HY-80 and HY-100 weldments, when at least 48 hours have elapsed after welding - and for other steel weldments, when weldments have cooled to room temperature after welding and stress relieving (when required for MIL-7018-M and MIL-10018-M1 test plates) - weld reinforcement and backing strip shall be removed flush with test plate surfaces.
(l) The weld shall be inspected radiographically according to level 2-2T as specified in MIL-STD-271 for conformance to 3.9.
(m) Two tensile specimens, when required, shall be stress-relieved in accordance with 3.4 herein and two tensile specimens shall be tested in as-welded condition.
(n) Tension test specimens shall be machined and tested in accordance with AWS B4.0. Tension testing shall be conducted at room temperature.
(o) For Charpy V-notch testing, specimens shall be machined and tested in accordance with AWS B4.0. Five specimens shall be required for each Charpy testing temperature shown in table III for the as-welded condition, and, when required (see 6.2), on the stress-relieved condition.
(p) Dynamic tear testing, if selected, shall have specimens machined and tested in accordance with ASTM E 604. Two specimens shall be required for each dynamic tear testing temperature specified in table III.

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4.5.2 Explosion test assemblies and mechanical properties prolongation tests. The two explosion test assemblies will be tested by NAVSEA as crack starters in accordance with MIL-STD-2149. The mechanical prolongations will be tested by NAVSEA in accordance with MIL-STD-2149.

4.5.2.1 Welding parameters. Fabrication of the explosion test assemblies and mechanical property prolongation assemblies shall be as follows:

(a) Base plate material shall be as follows: for MIL-10018-M1 weld metal testing HY-80, in accordance with MIL-S-16216 and for MIL-12018-M2 weld metal testing - HY-100, in accordance with MIL-S-16216.
(b) Dimensions of the test assemblies shall be in accordance with MIL-STD-2149.
(c) The weldments shall be fabricated in the flat position with 3/16-inch and larger diameter electrodes or in the vertical-up position with 5/32-inch and small diameter electrodes using dc, reverse polarity.
(d) The preheat and interpass temperature shall be 250 $\pm$ 25°F.
(e) The welding-heat input shall be 50,000 to 60,000 joules per inch for the 1/8-inch size and larger electrodes and 30,000 to 40,000 joules per inch for 3/32-inch size electrodes.
(f) Peening of weld beads shall not be permitted.
(g) Welding amperage shall be in accordance with the manufacture's recommendations and the actual values used shall be recorded and reported.
(h) The joint surfaces shall not be clad or buttered.
(i) Welding may be continuous except for interpass cooling. No post-weld heat treatments shall be employed. Time delay per pass beyond that necessary for interpass cooling shall be reported.

4.6 Diffusible hydrogen. A sample of every lot of each size and MIL type of electrodes furnished to the requirements of this specification shall be tested for diffusible hydrogen level in the weld metal. The test results shall be recorded along with the corresponding covering moisture. The hydrogen level of the deposited weld metal shall be determined by the analytical method of gas chromatography or by the method of collection over mercury as specified in AWS A4.3. The test data shall be reported as specified in AWS A4.3.

4.7 Moisture resistance test procedure. The electrode sample shall be exposed in a Blue M environmental chamber model FR 251-1 or equivalent for 9 hours minimum at 80°F minus 0, plus 5°F and 80 percent R.H. minus 0, plus 5 percent. The chamber shall have a two-pen recorder with the first pen for dry bulb temperature and the second pen for the differential between wet bulb

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and dry bulb temperatures. The chamber is available in all standard voltages for the various power supplies. The procedure shall be as follows:

(a) The electrode sample in unopened packages shall be heated to a temperature minus 0, plus 10°F above the dew point of the chamber at the time of loading.
(b) The electrode sample shall be loaded into the chamber without delay after the packages are opened.
(c) The electrodes shall be placed in the chamber in a vertical position on 1-inch centers.
(d) Time, temperature, and humidity shall be continuously recorded for the period that the electrodes are in the chamber.
(e) Counting of exposure time shall start when the required temperature and humidity in the chamber are established.
(f) At the end of the exposure time, the electrodes shall be removed from the chamber and a sample of the coating taken for moisture determination in accordance with MIL-E-22200.

4.7.1 Frequency of testing. The test of 4.7 shall be conducted for each lot except that after testing 10 consecutive lots with satisfactory test results, the frequency of testing can be reduced to one lot in very 10 lots. However, one unsatisfactory test result shall require reversion to the above cycle.

4.8 Identification, requirements, rejection, and retest. The identification of melts, heats, or lots; the general requirements for test specimens; the rejection of nonconforming lots or specimens; and the retest and resubmittal of nonconforming lots shall be in accordance with MIL-E-22200.

4.9 Certification/data report. Quality conformance test result data and wet mix equivalency test result data are shown in the sample forms on figures 3 and 4 (see 6.3).

4.10 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

5.2 Packaging. Packaging shall be as specified in MIL-E-22200.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. This specification is intended to cover several arc-welding steel electrodes of various alloys and strength levels that deposit weld metal that meets the mechanical property requirements given herein. These electrodes have an extremely low covering moisture content and should reduce the likelihood of underbead cracking in welding of such high yield strength steels as HY-80 and HY-100 of MIL-S-16216.

6.1.1 MIL-7018-M. This type is intended for general use in the welding of ordinary strength and high tensile steels such as grade DH-36 and grade EH-36T of MIL-S-22698 and for attachment of such steels to high yield strength steels such as HY-80 and HY-100. This type can be used for as-welded and stress-relieved applications. This electrode is intended for use with dc, reverse polarity electrical characteristics only.

6.1.2 MIL-10018-M1. This type is intended for general use in the welding of HY-80 and T-1 steels for as-welded and stress-relieved applications. This electrode is intended for use with dc, reverse polarity electrical characteristics only.

6.1.3 MIL-12018-M2. This type is intended for general use in the welding of HY-100 steel for as-welded applications. This electrode is intended for use with dc, reverse polarity electrical characteristics only.

6.1.4 All types with suffix RC. This type is intended to cover electrodes for welding pressure and strength welds that will be subject to irradiation.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number and date of this specification.
(b) Type, class, and size required (see 1.2).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Whether stress relieving treatments are required (see 3.4 and 4.5.1(n)).
(e) Whether the copper content is restricted as designated by the suffix RC to the MIL-type (see footnote 2 to table IV).
(f) Whether heat or lot identification is required for individual electrodes (see 3.6.1).
(g) For 3/32-inch size electrode, the length required (see footnote to table V).
(h) Whether stress relieved quality conformance inspection is required for types MIL-7018-M and MIL-10018-M1 (see footnote 2 to table VIII).
(i) Whether a second test weldment using grade A, ASTM A 710 or grade HSLA-80, MIL-S-24645 steel base plate material is required (see 4.5.1(b)).

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6.2.1 Electrodes should be ordered by the pound.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.9	DI-MISC-80678	Certification/data report	----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List No. 22200 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests must be made in accordance with "Provisions Government Qualification SD-6" (see 6.4.1).

6.4.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.5 Inspection after delivery. Post delivery inspection of electrodes to determine conformance to this specification and for acceptance thereof is the responsibility of the contracting activity.

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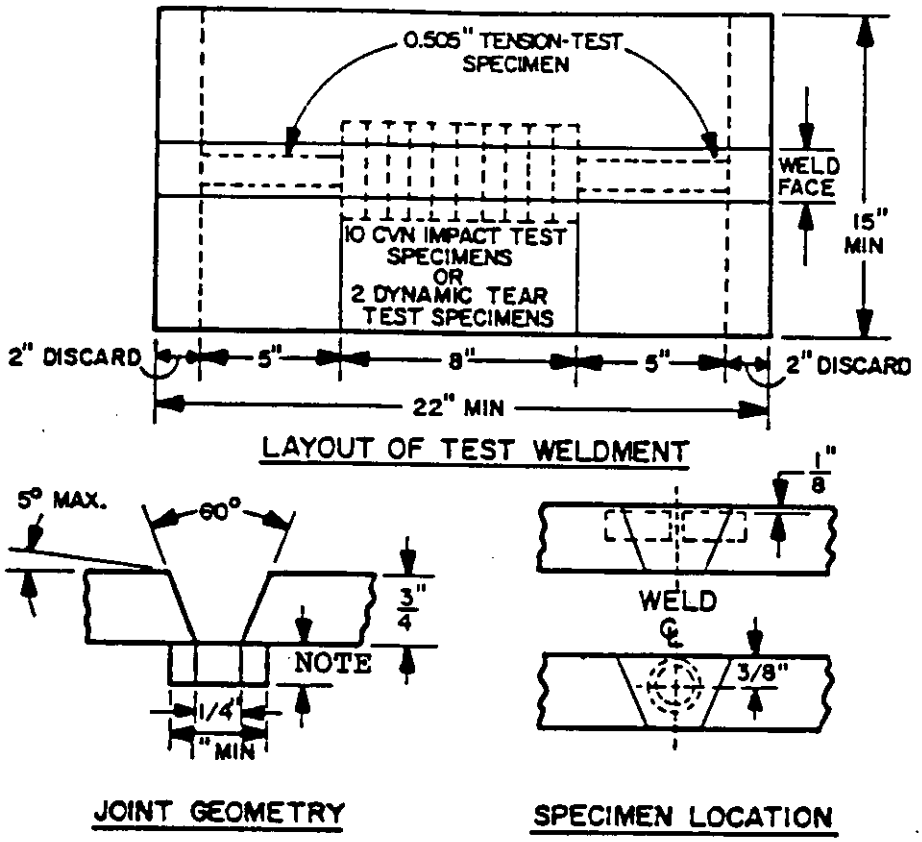
6.6 Subject term (key word) listing.

Dynamic tear
Elongation
HSLA-80 and -100 steel
HY-80, and -100 steel
Moisture resistant
Stress-relieved
Yield strength

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 3439-N677)

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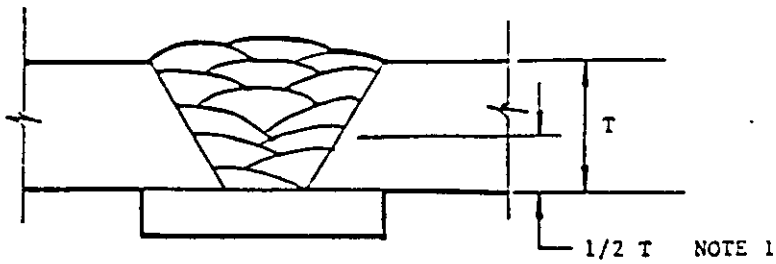


NOTE: See 4.5.1 (d) and (e).

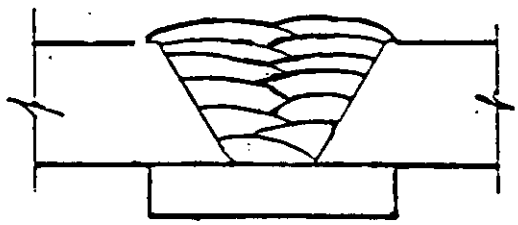
SH12319

FIGURE 1. Flat and vertical groove-weld metal test.

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BEAD SEQUENCE
(Acceptable)



UNDESIRED SPLIT WEAVE
(Unacceptable)

NOTE: The split-weave technique (the practice of depositing only two wide-weave beads per weld layer) shall be limited to the first 1/2T of weld joint thickness. Each weld layer beyond the first 1/2T of weld joint thickness shall contain three or more weld beads, as shown in the top figure.

FIGURE 2. Test weldment bead placement.

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CERTIFICATION OF QUALITY CONFORMANCE TESTS (SEE NOTES 1 AND 2

Manufacturer or distributor _____

Address _____

Date _____

Specification MIL-_____

Type MIL-_____

Diameter and length _____

Lot no. _____

Chemical analysis (Complete)

Carbon _____

Manganese _____

Silicon _____

Phosphorus _____

Sulfur _____

Chromium _____

Nickel _____

Molybdenum _____

Vanadium _____

Copper (see note 3). _____

Chemistry was taken from: Chem.pad___

Groove weld___

Customer's name _____

Customer's order no. _____

Core wire heat no. _____

Lot identification MIL-E-22200_____

Para. _____

Wet mix no. _____

Test plate condition _____

Mechanical test results

Yield strength _____ (0.2 percent offset method)

Tensile strength _____

Elongation _____

Reduction of area (percent) _____

Impact test results

Charpy V-notch ___ Dynamic tear ___

Testing temperature/Impact energy

1	_____/_____	1_____/_____
2	_____/_____	2_____/_____
3	_____/_____	3_____/_____
4	_____/_____	4_____/_____
5	_____/_____	5_____/_____

FIGURE 3. Certification of quality conformance tests.

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Concentricity (percent) _____

Covering moisture
Condition A _____
Condition B _____

Radiographic _____

Diffusible Hydrogen Data
(as specified in AWS A4.3)

Slag removal (see MIL-E-22200).
Grinding during test plate preparation.

Operator error (layer No.s)_____

Test/Analysis Method: Gas Chromatography ___, Over Mercury ___

We hereby certify that the above material has been tested in accordance with the listed specification and is in conformance with all requirements.

_____
Authorized contractor representative

Notes to figure 3:

1. Items not applicable to the type electrode involved or exempted by the acquisition document shall be marked N/A.
2. This form is applicable to the quality conformance tests required for each lot. Figure 4 forms shall be employed to report and certify the wet mix equivalency tests where applicable for each wet mix of covering mixture in the lot. One copy of figure 4 form shall be employed for certifying equivalency of each wet mix in a lot where applicable. One copy of the quality conformance certificate (figure 3) for the lot shall be forwarded with copies of the batch equivalency certificates for each additional mix from that lot that is included in the shipment.
3. Copper content is required for MIL-types with RC suffix (see footnote 2 to table IV).

FIGURE 3. Certification of quality conformance - Continued.

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CERTIFICATION OF WET MIX EQUIVALENCY TESTS (SEE NOTES 1 AND 2)

Manufacturer or distributor _____ Customer's name _____

Address _____

Date _____ Customer's order no. _____

Specification MIL- _____

Type MIL- _____

Diameter size _____ Core wire heat no. _____

Lot no. _____ Wet mix no. _____

Chemical analysis

Manganese _____

Nickel _____

Copper (see note 3) _____ Amperage _____

We hereby certify that the above material has been tested in accordance with the listed specification and is in conformance with all requirements.

_____
Authorized contractor representative

NOTES:

1. Items not applicable to the type electrode involved or exempted by the acquisition document shall be marked N/A.
2. One copy of this form shall be employed for certifying equivalency of each wet mix in a lot where applicable. One copy of the quality conformance certificate (figure 3) for the lot shall be forwarded with copies of the mix equivalency certificates for each additional mix from that lot that is included in the shipment.
3. Copper content is required for MIL-types with RC suffix (see footnote 2 to table IV).

FIGURE 4. Certification of wet mix equivalency tests.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-22200/10B(SH)	ELECTRODES, WELDING, MINERAL COVERED, IRON-POWDER, LOW-HYDROGEN MEDIUM, HIGH TENSILE AND HIGHER-STRENGTH LOW ALLOY

3a. NAME OF SUBMITTING ORGANIZATION STEELS

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional

b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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